

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	S. Mohammed Abrar	Reg. No.:	192472185
Section / Batch:	CSE-AI / 2024	Date:	21/07/2026

Code of Conduct

I S. Mohammed Abrar – [192472185] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Mohammed Abrar

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
5	22	19	20	15	10	86

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: T. Kumaragurubaran	Signature: _____	Signature: _____
Date: 21/07/26_____	Date: _____	Date: _____



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DSA0308 Natural Language Processing

MODELS AND ALGORITHMS IN NATURAL LANGUAGE PROCESSING (NLP)

Presented by

S Mohammed Abrar (192472185)

Guided by:

Dr. Kumaragurubaran. T



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INTRODUCTION



1. Natural Language Processing (**NLP**) is a branch of **Artificial Intelligence (AI)** that enables computers to understand, interpret, analyze, and generate human language.
2. It combines the fields of **Computer Science (CS)**, **Linguistics**, and **Machine Learning (ML)** to process text and speech efficiently.
3. Models and algorithms are the fundamental building blocks of NLP because they help computers learn language patterns, make predictions, and perform various language-related tasks accurately.



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WHAT ARE NLP MODELS?



An **NLP model** is a computational framework that is trained using large amounts of textual data to understand the structure and meaning of human language. These models learn linguistic patterns and relationships between words to perform specific NLP tasks.

Types of NLP Models

- **Rule-Based Models:** These models use manually designed linguistic rules and dictionaries to process text.
- **Statistical Models:** These models use probability and statistical techniques to predict the next word or classify text.



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WHAT ARE NLP MODELS?



- **Machine Learning (ML) Models:** These models learn patterns from labeled datasets without requiring explicit programming.
- **Deep Learning (DL) Models:** These models use artificial neural networks to capture complex relationships within language.
- **Transformer Models:** These are advanced deep learning models that use the attention mechanism to process entire sentences simultaneously.



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NLP ALGORITHMS



Naïve Bayes (NB)

- Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It is widely used for spam detection, sentiment analysis, and document classification because of its simplicity and efficiency.

Support Vector Machine (SVM)

- Support Vector Machine is a supervised learning algorithm that identifies the optimal decision boundary between different classes. It is commonly used for text categorization and sentiment analysis.

Hidden Markov Model (HMM)

- A Hidden Markov Model is a statistical sequence model that predicts hidden states from observable data. It is extensively used for **Part-of-Speech (POS)** tagging and speech recognition.



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NLP ALGORITHMS



Conditional Random Field (CRF)

- A Conditional Random Field is a probabilistic graphical model used for sequence labeling tasks. It is widely applied in **Named Entity Recognition (NER)** and information extraction.

K-Means Clustering

- K-Means is an unsupervised learning algorithm that groups similar documents into clusters without requiring labeled data.

Term Frequency–Inverse Document Frequency (TF-IDF)

- TF-IDF is a feature extraction technique that measures the importance of a word within a document relative to an entire collection of documents.

Word2Vec

- Word2Vec is a word embedding algorithm that converts words into numerical vectors while preserving their semantic relationships.



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OBJECTIVES



1. The primary objective of Natural Language Processing is to enable computers to understand and communicate using human language effectively.
2. NLP also aims to extract useful information from unstructured text, automate language-related tasks, improve decision-making, generate meaningful responses, and develop intelligent applications capable of interacting naturally with users.



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COMPARISON OF NLP MODELS



Model	Strength	Limitation
Rule-Based	Simple	Limited flexibility
Statistical	Learns probabilities	Requires quality data
Machine Learning	Good accuracy	Needs labeled data
Deep Learning	Learns complex patterns	Computationally expensive
Transformer	State-of-the-art performance	High training cost



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APPLICATIONS



- Natural Language Processing is widely applied across many real-world domains.
- Chatbots and virtual assistants use NLP to understand user queries and generate meaningful responses.
- Machine translation systems convert text from one language to another automatically.
- Sentiment analysis identifies positive, negative, or neutral opinions expressed in customer reviews and social media.
- Spam detection filters unwanted emails using text classification techniques.

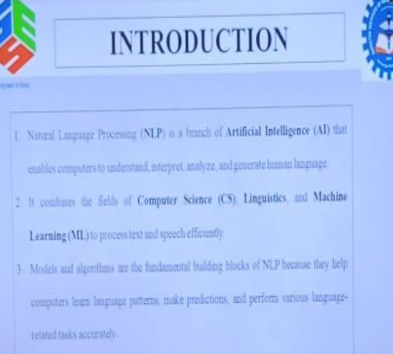


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CONCLUSION



- NLP models and algorithms form the foundation of intelligent language processing.
- Traditional machine learning methods remain useful for many applications, while deep learning and transformer models achieve state-of-the-art performance.
- The continued advancement of NLP is enabling more accurate, efficient, and human-like language understanding across industries.



INTRODUCTION

Dr. Jeyaraj K. Sivasankaran

1. Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, analyze, and generate human language.
2. It combines the fields of Computer Science (CS), Linguistics, and Machine Learning (ML) to process text and speech efficiently.
3. Models and algorithms are the fundamental building blocks of NLP because they help computers learn language patterns, make predictions, and perform various language-related tasks accurately.



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India

Lat 13.025916° Long 80.016721°

Tuesday, 14/07/2026 11:49 AM GMT +05:30



A man in a checkered shirt is standing next to a large screen displaying a presentation slide titled "INTRODUCTION". The slide lists three points about Natural Language Processing (NLP). To the left of the screen is a whiteboard with handwritten notes and diagrams. In the foreground, there is a small table with a laptop and other items. The bottom of the image features a Google Maps overlay showing the location of Mevalurkuppam, Tamil Nadu, India, with a red location pin and the Google logo.

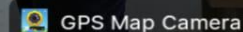


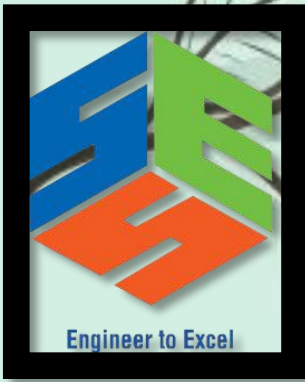
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Thank you